

Occupational exposure assessment using antibody levels: exposure to avian leukosis/sarcoma viruses in the poultry industry.

Avian leukosis/sarcoma viruses (ALSV) infect and cause cancers in chickens. Poultry workers are exposed to ALSV and other infectious agents in the workplace. This study examines if industrial hygiene assessment of antibody levels in poultry workers can identify risky job tasks at the higher exposure risk to an infectious agent, i.e., ALSV. We compared ALSV antibody levels in poultry workers and control subjects. Occupational and demographical factors were examined for an association with the exposure risk in poultry workers. We found that the antibody levels were significantly higher in poultry workers than in control subjects. Job category and age together were significantly associated with the antibody levels in workers. Certain job tasks were identified with significantly higher antibody levels as compared to others, implying that recommendations should be made to protect workers at these jobs. The findings of this study indicate that the measurement of antibody levels in workers can be useful for industrial hygiene assessment of exposure to infectious agents.